

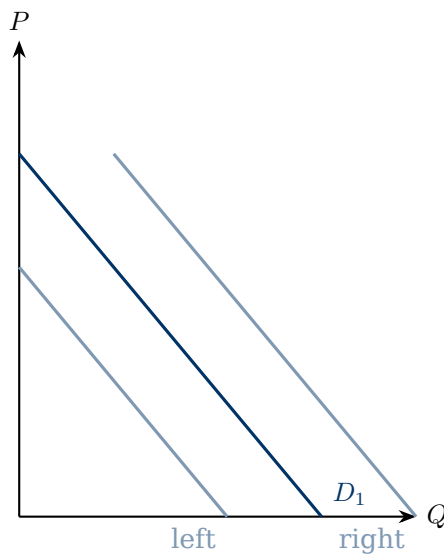
Handout 2: The Market Forces of Supply and Demand

Mankiw Ch 4

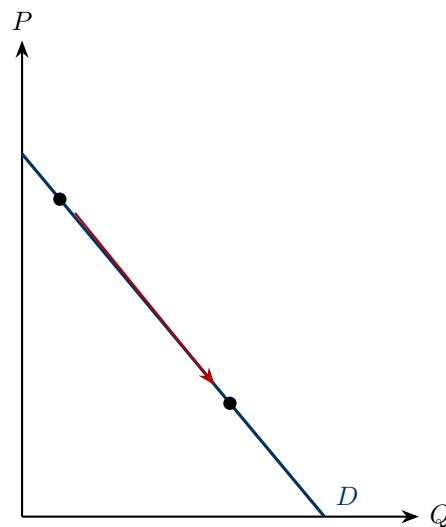
Answer key

Problem 1. Spotify and the demand curve

Spotify sells one subscription at one price in every country it serves. Draw the demand curve for subscriptions. For each event below, say whether it shifts the curve or moves you along it, say which direction, and give the reason in one line.



(a), (d), (e) right • (b), (c), (f) left



(g) movement along

- a) Spotify expands to twenty more countries without changing the price.

More buyers: market demand is the sum of individual demands, so the curve shifts **right**.

- b) A global recession raises unemployment and lowers incomes everywhere.

Income falls. If subscriptions are a normal good, demand shifts **left**. (If they were inferior, demand would rise; the honest answer names the assumption.)

- c) The price of Apple Music drops.

Apple Music is a **substitute**. Its price fell, so buyers move to it and demand for Spotify shifts **left**.

- d) The price of cell phones falls.

Phones are a **complement**: you need one to stream. A cheaper complement

shifts demand for Spotify **right**.

- e) Spotify starts promoting local artists from developing countries.

Tastes. The service becomes more attractive at any price, so demand shifts **right**.

- f) Prominent economists warn of a coming recession.

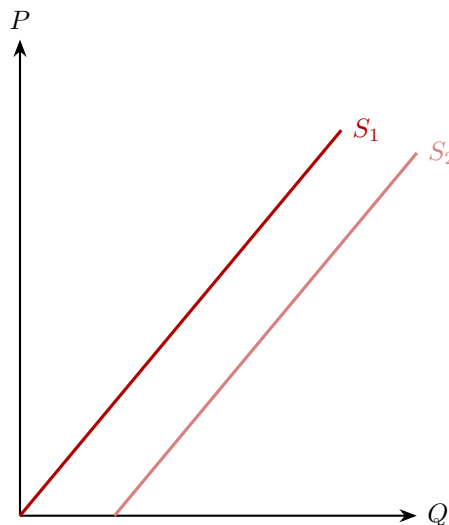
Expectations about future income. Buyers expecting lower income cut back today, so demand shifts **left**. Note that nothing has happened to income yet: the expectation alone moves the curve.

- g) Spotify ends its free tier, so new users must pay the subscription price.

This is the good's **own price** for those users, rising from zero to the subscription price. That is a **movement along** the demand curve, not a shift. The curve does not move.

Problem 2. Supply of electric cars: shift or movement?

Draw the supply curve for electric cars. For each event, say whether supply shifts, and in which direction, or whether something else happens.



(a) supply right

- a) The price of lithium, used in the batteries, falls.

An **input price** fell, so each car is cheaper to build and more are offered at every price: supply shifts **right**.

- b) The price of electric cars falls.

The good's **own price**. That is a **movement down along** the supply curve to a lower quantity supplied. The curve itself does not move.

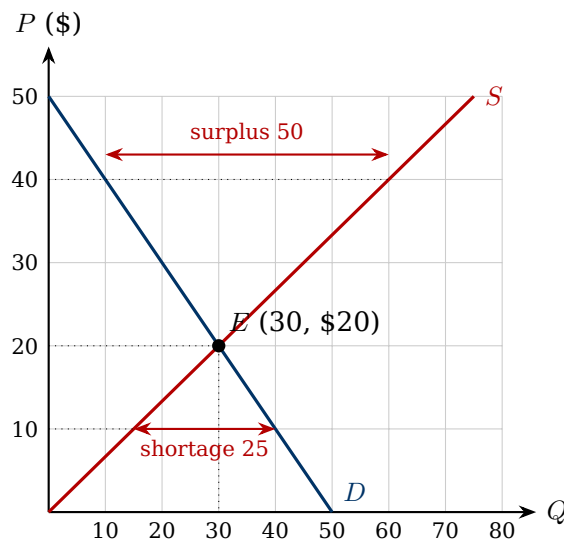
- c) The price of gasoline cars falls.

A trap. Gasoline cars are a substitute *for buyers*, not an input for sellers. Nothing about the cost of building an electric car has changed, so supply does not move. This shifts the **demand** for electric cars left.

Problem 3. Equilibrium from a schedule

The table gives the market demand and supply for a good.

P (\$)	Q_d	Q_s
0	50	0
10	40	15
20	30	30
30	20	45
40	10	60
50	0	75



- a) Plot both curves and find the equilibrium price and quantity.

The two quantities are equal at \$20, where $Q_d = Q_s = 30$. So $P^* = \$20$ and $Q^* = 30$: the curves cross at E .

- b) Find the surplus or shortage at $P = 40$, and say what happens next.

$Q_s = 60$ and $Q_d = 10$, a **surplus** of $60 - 10 = 50$. Sellers hold unsold stock and cut the price. As the price falls, Q_d rises along D and Q_s falls along S , and the surplus closes at \$20.

- c) Find the surplus or shortage at $P = 10$, and say what happens next.

$Q_d = 40$ and $Q_s = 15$, a **shortage** of $40 - 15 = 25$. Buyers compete for too few units and sellers raise the price. Q_d falls along D , Q_s rises along S , and the shortage closes at \$20.

- d) Write the two curves as equations.

Demand falls by 10 units for every \$10, from 50 at a price of zero: $Q_d = 50 - P$. Supply rises by 15 units for every \$10 from the origin: $Q_s = 1.5P$. Check: $50 - P = 1.5P$ gives $P = 20$ and $Q = 30$.

Problem 4. Equilibrium from equations

Demand and supply for a product are

$$Q_d = 1200 - 50P \qquad Q_s = 200 + 50P$$

- a) Calculate the equilibrium price and quantity.

Set $Q_d = Q_s$: $1200 - 50P = 200 + 50P$, so $1000 = 100P$ and $P^* = \$10$. Then $Q^* = 1200 - 50(10) = 700$. Check with the other equation: $200 + 50(10) = 700$. ✓

- b) Is there a surplus or a shortage at $P = 14$? How large?

$Q_d = 1200 - 700 = 500$ and $Q_s = 200 + 700 = 900$. A **surplus** of 400. The price is above equilibrium, so it will be bid down toward \$10.

- c) Buyers' incomes rise and demand becomes $Q_d = 1500 - 50P$. Find the new equilibrium.

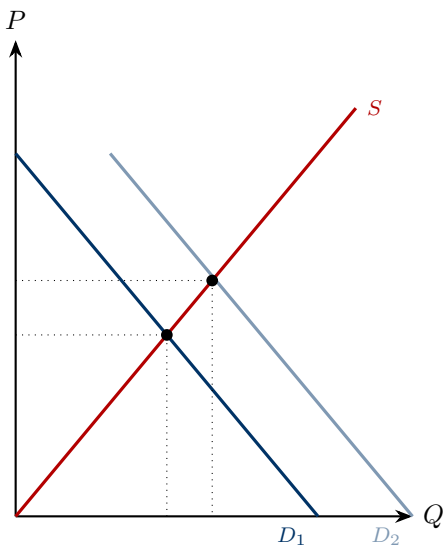
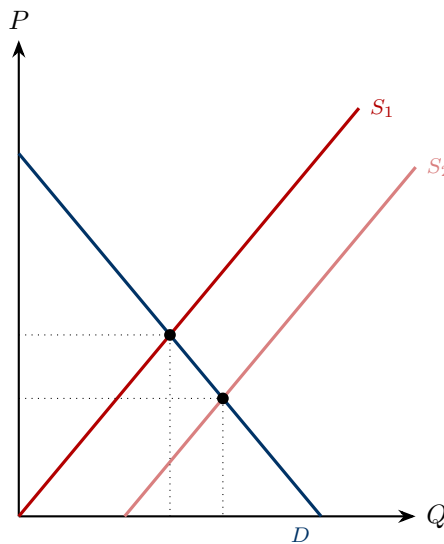
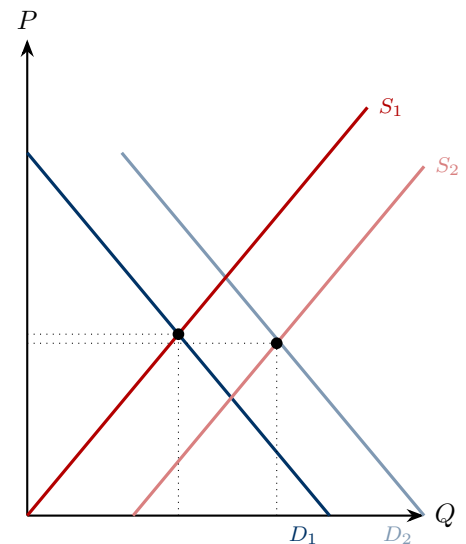
$1500 - 50P = 200 + 50P$, so $1300 = 100P$ and $P^* = \$13$. Then $Q^* = 1500 - 50(13) = 850$. Both price and quantity rise, which is what an increase in demand against an upward-sloping supply curve always does. The good is normal, since higher income raised demand.

- d) Immediately after the shift, the price is still \$10. What does the market look like at that price, and how does it get to the new equilibrium?

At \$10: $Q_d = 1500 - 500 = 1000$ and $Q_s = 200 + 500 = 700$, a **shortage** of 300. That shortage is the mechanism: buyers bid against each other, the price rises, Q_d falls along the new demand curve and Q_s rises along supply, until the gap closes at \$13.

Problem 5. The electric car market in equilibrium

The market for electric cars starts in equilibrium. For each event, use the three steps: which curve shifts, in which direction, and what happens to equilibrium price and quantity.

(a) $P \uparrow, Q \uparrow$ (b) $P \downarrow, Q \uparrow$ (c) $Q \uparrow, P$ **ambiguous**

- a) The price of oil rises.

Gasoline cars are a **substitute** and have just become costlier to run, so buyers want more electric cars at any price. Nothing has changed about building one, so this is demand, not supply. **Demand shifts right.** At the old price there is a shortage, the price is bid up, and both P and Q rise.

- b) New technology reduces the cost of engine design for electric cars.

This is a **cost** of production, so it works on supply. **Supply shifts right.** At the old price there is a surplus, sellers cut the price, so P falls and Q rises.

- c) The oil shock and the technology shock happen at the same time.

Both curves shift right. Quantity rises in both cases, so Q **rises for certain**. The two shifts push price in opposite directions, so P **is ambiguous**: if the demand shift is larger the price rises, if the supply shift is larger it falls. Nothing in the question says which is larger, so the honest answer is ambiguous, not a guess.

Problem 6. Three scenarios

Use the same three steps. Name the shifter, not just the direction.

- a) A hurricane destroys much of the Hudson Valley apple crop. What happens in the market for apples?

Fewer apples can be brought to market at any price. **Supply shifts left** (the shifter is the loss of a productive input). P rises and Q falls. This is roughly what happened after Hurricane Irene in 2011.

- b) A report finds dangerous mercury levels in fish. In the same month, the diesel fuel that fishing boats burn gets cheaper. What happens in the market for fish?

Two shifts. The report works on **tastes**, so demand shifts left. Cheaper diesel is an **input price**, so supply shifts right. Both push the price **down**, so P falls for certain. They pull quantity in opposite directions, so Q **is ambiguous**.

- c) Honey and jam are substitutes, and the price of honey rises. What happens in the market for jam? What happens in the market for honey?

Jam: the price of a **substitute** rose, so buyers move to jam. Demand for jam shifts right and both P and Q of jam rise.

Honey: the event is honey's **own price**, so there is no shift in the honey market. It is a **movement along** the honey demand curve to a lower quantity demanded.

Watch for this: the same event is a shifter in one market and a movement along the curve in the other.